

Multivariable Calculus

By now, you're familiar with calculus involving functions of a single variable. In reality, though, many applications involve functions of multiple variables, $f(x, y, z)$. What does it mean to do limits, derivatives, and integrals involving such functions?

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1. Higher-Dimensional Spaces (TODO)

In case you're entirely unfamiliar with the concepts involving 3d+ spaces, this will be a quick rundown.

2. Multivariable Limits

Just as limits help us understand derivatives and infinitesimals in singlevariable calculus, it is useful to begin with multivariable limits here as well. Understanding the differences between multivariable and singlevariable limits will help build an intuition for the differences between multivariable and singlevariable calculus as a whole.

2.1. The Concept of Multivariable Limits

The limit of $g(x)$ as $x \rightarrow a$ is the value that g approaches as the distance between x and a decreases on the domain of g , a 1d number line.

Similarly, the limit of $f(x, y)$ as $(x, y) \rightarrow (a, b)$ is the value that f approaches as the distance between the points (x, y) and (a, b) decreases on the domain of f , a 2d plane. And so forth for higher dimensions.

(TODO) visual

This leads to some terminology. A function of two variables $f(x, y)$ is graphed in 3d space as $(x, y, f(x, y))$. Its limit, however, is referred to as "2 dimensional" because the limit operates over f 's domain, (x, y) . It follows, then, that a function of 1 variable has a 1d limit, or that a function of n variables has a nd limit.

Now, with 1d limits, recall the concept of one-sided limits. Because we are on a 1d line, x can only approach a from either the left or the right. A “true” limit, then, only exists if both the left and right limits agree. We will now generalize this concept to higher dimensions.

Let us consider the limit of $f(x, y)$ as $(x, y) \rightarrow (a, b)$. The points (x, y) and (a, b) are on a 2d plane, and so (x, y) can theoretically approach (a, b) from infinite, straight-line directions, as well as along any curved path which passes through (a, b) .

(TODO) visual

In order to prove that a limit to exist, we must show that *all paths* agree. With 1d limits, this was trivial since there were only 2 approaches, the left and right. With 2d+ limits, we must show that this is true for all of the infinite possible paths.

(TODO) notation, techniques for solving, factoring out a var, grouping and substituting, parametrics, higher-dimensions, a generalized approach?, iterated limits, squeeze theorem techniques

3. Multivariable Derivatives

What does it mean to take the derivative of a multivariate function? Before calculating such derivatives, let us first try and build an intuitive conceptual understanding of how univariate derivatives generalize into higher dimensions.

3.1. A Geometric Introduction

Geometrically, the derivative is a slope. It’s easy to see that a function of one variable has a single slope at any point. It’s a line, and so the slope is the direction in which the line is pointing.

A 3d function, though, isn’t a line; it’s a surface. And so, what is the “slope” of a surface? Theoretically, from some point on the surface, we could draw a line in any direction and have a different slope. There are infinite “slopes” in the same way that there were infinite limit-directions. So, at least in some sense, we might expect there to be infinite derivative-like values associated with some point. Each of these is called a “directional derivative”, and they will be discussed below.

(TODO) visual

Now, let us consider tangents. Recall that we define the slope of a curve at some point to be the slope of the tangent to the curve at that point. If a surface has infinite slopes at some point, we could expect it to also have infinite tangents at that point.

(TODO) visual

Notice that the tangents, though, are coplanar with each other. And so, considering all of the tangents at once, we can see that a 2d function has a single tangent *plane* at any point. And so, we can also relate the derivative of a multivariate function to some *single* concept rather than infinite slopes.

(TODO) visual

Now, in order to describe a plane, it is sufficient to have two vectors from within that plane. So, in theory, we need only two of our directional derivatives in order to describe the entire tangent plane. These particular derivatives are called “partial derivatives”, and they will be discussed next.

Finally, extrapolating into higher dimensions, we might correctly guess that some point on an nd volume has an nd tangent volume. Therefore, we would need n partial derivatives in order to describe the tangent to a function of n variables. (Notice that this is consistent with our understanding of a function of 1 variable, as well.)

3.2. Partial Derivatives

4. Application: Approximation Using Partial Derivatives

5. Application: Tangent Planes

6. Multivariable Extreme Values

7. Application: Extreme Values Under Constraints

7.1. Lagrange Multipliers

8. Integrals in Higher Dimensions

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